Freight Rate Prediction: Approach & Validation Report

1. Data overview

48,000 labeled loads from 2025-01-01 to 2025-10-31 across 64 cities and 3 equipment types. Rate is driven mainly by distance (correlation 0.91). Median $/mile by equipment: Dry Van 2.046, Flatbed 2.22, Reefer 2.312. Median rates rise from about $1,915 in January to a June peak of about $2,120, then level off.

2. Data-quality issues and fixes

• **Negative weights** (292 rows): magnitudes look like normal loads, so I treated them as sign errors and took the absolute value.
• **Missing weight** (300): filled with the training median.
• **Weight capped at 47,500 lb** (1204 rows): kept and clipped consistently.
• **Missing market_index** (374): left as missing; LightGBM handles NaN natively.
• **Rate outliers**: about 1.3% of rows have $/mile 2–7x above or below their lane's median, likely entry or unit errors. I removed 628 training rows outside 0.5x–2x of their lane + equipment median. Prediction rows are never dropped.

3. Split and validation approach

The final predictions cover Nov–Dec 2025, which are future dates the model has never seen. A random split would let the model see market conditions from the same weeks it is tested on and overstate accuracy. I used a **time-based holdout**: train 2025-01-01..2025-08-31 (37983 rows) | test 2025-09-01..2025-10-31 (9389 rows). After choosing the model, I retrained it on all cleaned data (Jan–Oct) to produce the final predictions.

4. Model

Target: log(rate / distance). Rate per mile makes short and long hauls comparable, and the log reduces skew. Features: distance, straight-line miles from coordinates, lat/lon, pickup/delivery/equipment as categoricals, weight, market_index, quote_signal, and calendar fields (day of year, day of week, month). Model: LightGBM gradient boosting (800 trees, learning rate 0.05, 31 leaves), chosen because it handles mixed categorical and numeric data, missing values and non-linear interactions with little tuning. Baseline: median $/mile for each lane + equipment, times distance.

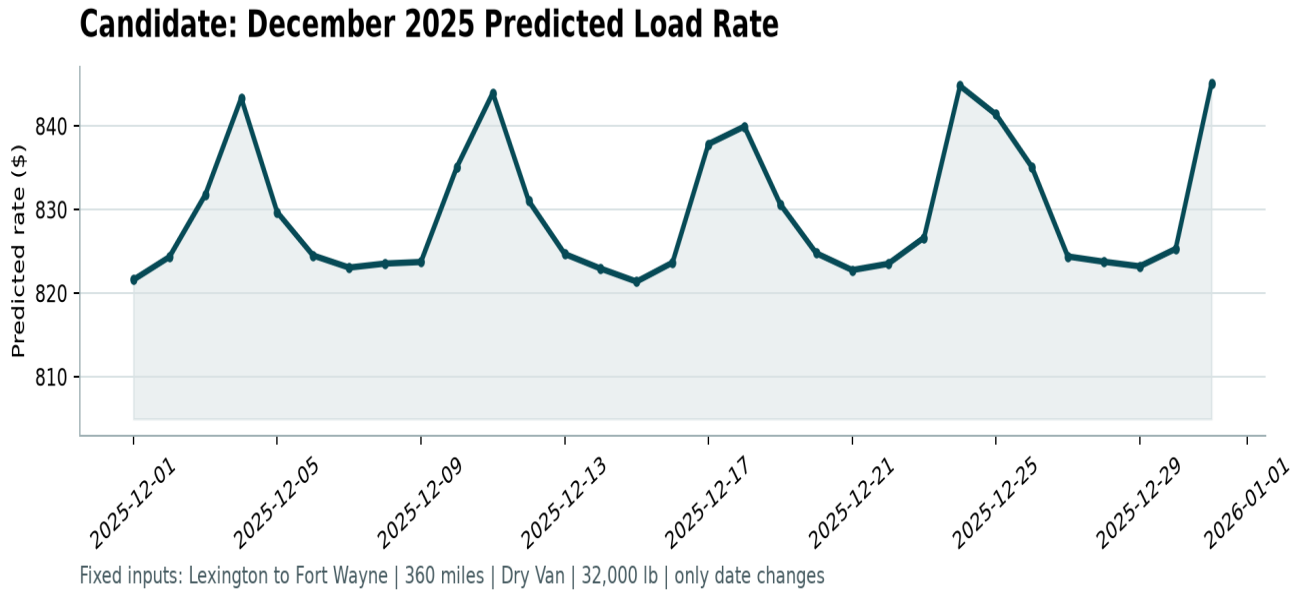
5. Results (Sep–Oct holdout)

Model / holdout	MAE $	RMSE $	MAPE
Baseline (cleaned)	93.6	218.7	3.89%
LightGBM (cleaned)	58.7	189.2	2.30%
Baseline (raw, incl. outliers)	145.6	651.7	6.33%
LightGBM (raw, incl. outliers)	109.7	634.8	4.68%

LightGBM cuts MAE by about 37% against the baseline. The raw-holdout numbers include the suspected erroneous rows, which no model can predict. Top features by gain: distance 26.9%, great_circle_miles 26.3%, equipment 15.3%, quote_signal 13.1%, pickup 3.2%.

6. December prediction chart (produced by score.py)

Inputs are fixed (Lexington to Fort Wayne, 360 mi, Dry Van, 32,000 lb); only the date changes. The chart file has no market_index, quote_signal or coordinates. Coordinates come from the city lookup, and market_index and quote_signal are the daily median for that date in validation.csv, so no values are invented. Predictions stay in a narrow $821–$845 band with a weekly pattern.



7. Limitations

Ten months of history cannot show a full-year seasonal pattern, so December seasonality is approximated by the most recent behavior. With more time I would add lane-level rolling rate features and quantile models for price ranges.